HPSC SCREENING TEST FOR ASSISTANT PROFESSOR 2025

1. Given below is a statement and two options :
Let $V = \mathbb{R}^2(\mathbb{R})$ and $u = (1, -1)$. Let $W = \{(x, y) : y = 2x\}$ be a subspace of V. Then the 90° rotation of the orthogonal projection of u on W of the inner product space V is:

Option I: $\left(\frac{2}{\sqrt{5}}, \frac{-1}{\sqrt{5}}\right)$

Option II : $\left(\frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$

In the light of the above statements, choose the correct answer from the options given below :
(A) Both option I and option II are true.
(B) Both option I and option II are false.
(C) Option I is correct but option II is false.
(D) Option I is incorrect but option II is true.
(E) Question not attempted

2. Consider the following statement and the options given below :
Let $A \in M_{2\times 2}(\mathbb{R})$ such that $A^2 = 0$. Then

1. A is similar to $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$,
2. A is similar to $\begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$, where $A \neq 0$.
3. A is similar to $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ only if $A \neq 0$.

How many of the options given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

3. Consider the following statements :
1. Let $f : \mathbb{R}^2 \to \mathbb{R}$ by $f(x, y) = x^{[y]}$, then for every $x \in \mathbb{R}, y \to f(x, y)$ is continuous on $\mathbb{R}$, where $[y]$ is the greatest integer function.

2. There exists a continuous function $f : \mathbb{R} \to \mathbb{R}$ that takes every value thrice.
3. There exists a continuous function $f : \mathbb{R} \to \mathbb{R}$ that takes every value twice. How many of the statements given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

4. Consider the following statements :

1. If A and B are closed subsets of $\mathbb{R}$, then the Minkowski sum $A + B$ is also closed in $\mathbb{R}$.

2. If A and B are connected subsets of $\mathbb{R}$, then the Minkowski sum $A + B$ is also connected.

3. Let $a, b \in \mathbb{R}$ with $a < b$, let $f : (a, b) \to \mathbb{R}$ be a continous function. Then f is uniformly continuous if and only if it is bounded.

How many of the statements given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

5. Given below are two statements :

Statement I : Let $f \in C^1[0, 1]$ satisfying $f(0) = f(1) = 0$, then
$\max_{0 \le x \le 1} |f'(x)| \ge 4 \int_0^1 |f(x)|dx$
Statement II : Let $g \in C'[0, 1]$ satisfying $g(0) = g(1) = 0$, then
$\max_{0 \le x \le 1} |g'(x)| \ge T \int_0^1 |g(x)|dx$, where $T > 4$.
In the light of the above statements, choose the correct answer from the options given below :
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is correct but Statement II is false
(D) Statement I is incorrect but Statement II is true
(E) Question not attempted

6. Consider the following statement and the options given below :
Let $W(\mathbb{F}_1)$ and $V(\mathbb{F}_2)$ be two vector spaces. Then a map $T : W(\mathbb{F}_1) \to V(\mathbb{F}_2)$ becomes a linear transformation if :
1. For any choices of $\mathbb{F}_1$ and $\mathbb{F}_2$
2. If one of them contains the other
3. If $\mathbb{F}_1 \subseteq \mathbb{F}_2$
4. If $\mathbb{F}_1 = \mathbb{F}_2$

How many of the options given above are correct?
(A) Only one
(B) Only two
(C) Only three
(D) All four
(E) Question not attempted

7. Given below are two statements :

Statement I : Let $S = \{(1,1,0),(1,0,2),(0,1,1)\}$ be a subset of the vector space $\mathbb{F}^3(\mathbb{F})$. Then S is linearly independent set for any choices of the field $\mathbb{F}$.

Statement II : The set S defined in I is a basis of V if $\mathbb{F} = \mathbb{R}$.

In the light of the above statements, choose the correct answer from the options given below :
(A) Both statement I and statement II are true.
(B) Both statement I and statement II are false.
(C) Statement I is correct but statement II is false.
(D) Statement I is incorrect but statement II is true.
(E) Question not attempted

8. Given below are two statements :

Statement I : Lex X be a countable set. There exists a metric d on X such that (X, d) is complete.

Statement II : Lex X be a countable set. Then there exist a metric d on X such that (X, d) is compact.

In the light of the above statements, choose the correct answer from the options given below :
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is correct but Statement II is false
(D) Statement I is incorrect but Statement II is true
(E) Question not attempted

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9. Consider the following pairs :

1. C[0, 1] with $\|f\|_\infty$: It is complete
2. C[0, 3] with the norm $\|f\|_\infty + |f(1/2)|$: It is complete
3. C[0, 4] with $\|f\|_2$: It is complete

How many of the above pairs are correctly matched?
(A) Only one pair
(B) Only two pairs
(C) All three pair
(D) None of the pairs
(E) Question not attempted

10. Consider the following statement and the options given below :

If λ is an eigenvalue of $A = \begin{bmatrix} 1 & 2+4i & 5i & 1 \\ i & 3+4i & 6i & 1 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$, then

1. $|\lambda| \leq 7 + \sqrt{34}$
2. $|\lambda| \leq 6 + 2\sqrt{5}$
3. $|\lambda| \leq 1 + 3\sqrt{5} + \sqrt{34}$
4. $|\lambda| \leq 13$

How many of the options given above are correct?
(A) Only one
(B) Only two
(C) Only three
(D) All four
(E) Question not attempted

11. Consider the following statement and the options given below :

Let $A \in GL_n(\mathbb{R})$ such that $\text{adj}(\text{adj}(\text{adj}\, A)) = \begin{bmatrix} 2 & 4 & 0 \\ 0 & 128 & 0 \\ 100 & 3 & 1 \end{bmatrix}$.

Let $f(\lambda) = \lambda^3 + a\lambda^2 + b\lambda + C = 0 \in \mathbb{Z}[A]$ be the characteristic equation of A. Then $\mathbf{C} =$

1. 2
2. -2

3. $\frac{24}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{2n^2}$

4. $-\frac{24}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{2n^2}$

How many of the options given above are correct?
(A) Only one
(B) Only two
(C) Only three
(D) All four
(E) Question not attempted

12. Consider the following statement and the options given below : Let $\lambda_1 = 1$ and $\lambda_2 = 2$ are two eigenvalues of a matrix $A \in M_{2\times 2}(\mathbb{R})$. Then $\det(\text{adj } A) = |\text{adj } A|$ and trace (adj A); respectively.

1. $2\sum_{n=1}^{\infty}\frac{(-1)^{n-1}(2\pi)^{2n}}{(2n)!}, 3$
2. $-2, -3$
3. $2, 3\sum_{n=1}^{\infty}\frac{\left(\frac{\pi}{2}\right)^{2n-1}(-1)^{n-1}}{(2n-1)!}$
4. $-1, -3$

How many of the options given above are correct?
(A) Only one
(B) Only two
(C) Only three
(D) All four
(E) Question not attempted

13. Consider the following statements :
1. A and B are positive definite matrices iff $A + B$ is positive definite always.
2. If A is negative definite matrix, then A^k is negative definite iff k is odd.
3. Let A is negative definite matrix, then A^k is positive definite iff k is even.

How many of the statements given above are correct?
(A) Only one
(B) Only two
(C) all three
(D) None
(E) Question not attempted

14. Given below are two statements :

Let A be a semi-positive definite symmetric matrix and $AB + BA = 0$ for same $B \in \mathbb{C}_{n\times n}(\mathbb{C})$. Then

Statement I: AB and BA are nilpotent matrices of index one.

Statement II: AB and BA are nilpotent matrices of index two.

In the light of the above statements, choose the correct answer from the options given below :
(A) Both statement I and statement II are true.
(B) Both statement I and statement II are false.
(C) Statement I is correct but statement II is false.
(D) Statement I is incorrect but statement II is true.
(E) Question not attempted

15. Consider the following pairs :

1. $P_2(\mathbb{R})/\mathbb{R}$: dimension 2
2. $M_2(\mathrm{C})/\mathbb{R}$: dimension 4
3. $M_2(\mathbb{R})/\mathbb{Q}$: dimension infinite

How many of the above pairs are correctly matched?
(A) Only one pair
(B) Only two pairs
(C) All three pairs
(D) None of the pairs
(E) Question not attempted

16. Given below are two statements :
Statement I : Two matrices have same characteristics polynomials are similar.
Statement II : Two matrices have same minimal polynomial are similar.
In the light of the above statements, choose the correct answer from the options given below :
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is correct But Statement II is false
(D) Statement I is incorrect but Statement II is true
(E) Question not attempted

17. Consider the following statement and the options given below :

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Let $A = [e_2\ e_3\ e_4\ e_1\ e_6\ e_7\ e_5]$, where $e_i = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{1\times 7}$, 1 in the i-th position.

Then the possible $k \in \mathbb{N}$ such that $A^k = I$ is :
1. 7
2. 10
3. 12
4. 24

How many of the options given above are correct?
(A) Only one
(B) Only two
(C) Only three
(D) All four
(E) Question not attempted

18. Given below are two statements :

Let λ be an eigenvalue of $A = \begin{bmatrix} 1 & 2+4i & 5 & i \\ 1 & 3+4i & 6 & i \\ 0 & i & -1 & 0 \\ 1 & 0 & 0 & i \end{bmatrix}$ then

Statement I : There exists an eigenvalue of λ of A such that $12 < |\lambda| \leq 13$.

Statement II : If λ is an eigenvalue of A then $|\lambda| \leq 12$.

In the light of the above statements, choose the correct answer from the options given below :
(A) Both statement I and statement II are true.
(B) Both statement I and spatement II are false.
(C) Statement I is correct but statement II is false.
(D) Statement I is incorrectrout statement II is true.
(E) Question not attempted

19. Consider the following statement and the options given below :

The absolute value of the integral $\int_0^{\pi/2} \frac{x\left(1+x^2\right)}{(2i+2x)^2} \sin x dx, x \neq i$ is bounded by :

1. $\pi^{3/2}/16\sqrt{6}$
2. $\pi^2/16\sqrt{6}$
3. $\pi^2/8\sqrt{6}$
4. $\pi/8\sqrt{6}$

How many of the options given above are correct?
(A) Only one
(B) Only two
(C) Only three
(D) All four
(E) Question not attempted

20. Consider the following statement and the options given below : Let $A_{n\times n}$ and $B_{n\times n}$ be two idempotent matrices. Then :

1. If Nullity (A) = Nullity (B), then A is similar to B.
2. If Rank (A) = Rank(B), then A is similar to B.
3. 1 and 2 both are correct.

How many of the options given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

21. Consider the following statement and the options given below :

Let $A = \begin{bmatrix} 1 & 5 & 6 \\ 11 & a & 0 \\ 2 & -1 & 5 \end{bmatrix}$, then

1. A is positive definite for some $a \in \mathbb{R}$.
2. A is negative definite for some $a \in \mathbb{R}$.
3. A is indefinite if $7a < -341$.

How many of the options given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

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22. Consider the following statement and the options given below : Let $A \in M_{n\times n}(\mathbb{R})$ with characteristic polynomial $f(x) = (x-1)^k g(x)$, where

$$k = 2025 \lim_{n\to\infty} \exp\left(\sum_{t=1}^{n} \frac{1}{n+t}\right)$$

1. If A is diagonalizable, then $\dim(E_{\lambda=1}) = 2025$, for $g(1) \neq 0$.
2. If A is diagonalizable, then $\dim(E_{\lambda=1}) = 4050$, for $g(1) \neq 0$.
3. If A is diagonalizable, then $\dim(E_{\lambda=1}) = \frac{48600}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^2}$, for $g(1) \neq 0$.
4. If A is diagonalizable, then $\dim(E_{\lambda=1}) = 6075$, for $g(1) \neq 0$.

How many of the options given above are correct?
(A) Only one
(B) Only two
(C) Only three
(D) All four
(E) Question not attempted

23. Consider the following statement and the options given below : If $q(x,y) = x^2 + 2xy + 2y^2 - 3z^2$ be the given quadratic form, then its rank, index and signature are, respectively :
1. $1, 2, 3$
2. $3, 2, 1$
3. $3, 1, -1$

How many of the options given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

24. Consider the following statement and the options given below :

Let $A = \begin{bmatrix} 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & -2 \end{bmatrix}$, then its eigenvalues are :

1. $1, 1, -4 \pm \sqrt{2}$
2. $1, 1, 2 \pm \sqrt{2}$

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3. $1, -1, -2 \pm \sqrt{2}$

How many of the options given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

25. Consider the following statement and the options given below : Let $A \in \mathbb{C}^{m\times n}, B \in \mathbb{C}^{n\times p}, C \in \mathbb{C}^{p\times q}$ and $D \in \mathbb{C}^{q\times r}$, then which of the following inequalities holds:

1. $\rho(ABCD) + \rho(C) \geq \rho(ABC) + \rho(CD), \rho(A) = \text{rank}(A)$
2. $\rho(ABCD) + \rho(BC) \geq \rho(ABC) + \rho(BCD)$
3. $\rho(ABCD) + q \leq \rho(ABC) + \rho(D)$

How many of the options given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

26. Consider the following statement and the options given below :
Let $T : \mathbb{R}^2(\mathbb{R}) \to \mathbb{R}^2(\mathbb{R})$ be a linear transformation defined as $T(x, y) = (y, -x)$. Then
1. Eigenvalues of T are $\pm i$.
2. Eigenvalues of T are ± 1.
3. T has no eigenvalues.

How many of the options given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

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27. Given below are two statements :

Let H be a 11 -Sylow subgroup of the group $GL_{33}(\mathbb{F}_{11})$ of invertible matrices 33×33 matrices with entries from the field $\mathbb{F}_{11}$.

Statement I : Let $O(H) = k$ and $k \equiv t(\text{mod}13)$, then $t = 11$.

Statement II : Let $O(H) = k$ and $k \equiv t(\text{mod}13)$, then $t = 1$.

In the light of the above statements, choose the Correct answer from the options given below :
(A) Both Statement I and §tatement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is correct but Statement II is false
(D) Statement I is incorrect but Statement II is true
(E) Question not attempted

28. Consider the following statements :
Let $U(n) = \{m : 1 \leq m < n, (m, n) = 1, n > 1\}$.
1. Number of elements of order 4 in $U(32)$ are 2 .
2. $U(8) \cong V_4 =$ Klein 4-group.
3. $0(U(128)) = 64$.
How many of the statements given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

29. Consider the following statements :
Locate the zeroes for $f(z) = z^9 - 8z^2 + 5$.
1. No zeroes lying inside circle $|z| = \frac{1}{2}$.
2. Two zeroes lie within $\frac{1}{2} \leq |z| < 1$.
3. Seven zeroes lie within $1 \leq |z| < \frac{3}{2}$.
How many of the statements given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

30. Consider the following statements : Let $f(z) = u(x, y) + iv(x, y)$ be entire function.
1. If $u(x, y) \leq x \ \forall \ z = x + iy$ then $f(z)$ is a polynomial of degree atmost one.
2. If $v(x, y) \geq x \ \forall \ z = x + iy$ then $f(z)$ is a polynomial of degree 1 .
3. If $f(z)$ is non-constant then $f(\mathbb{C})$ is dense in $\mathbb{C}$.
How many of the statements given above are correct?
(A) Only one
(B) Only two
(C) All three

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(D) None
(E) Question not attempted

31. Consider the following Statements :
1. If all the zeroes of a polynomial lie in a half plane, then zeroes of derivative also lie in the same half plane.
2. If all zeroes of a polynomial lie in the unit circle then zeroes of derivative lie in closed unit disk $\{z \in \mathbb{C} : |z| \leq 11\}$.
3. If all zeroes of a polynomial are reals then zeroes of derivative are also reals.
4. If all zeroes of a polynomial are purely imaginary then zeroes of derivation are also purely imaginary.
How many of the statement given above are correct?
(A) Only one statement
(B) Only two statements
(C) All three statements
(D) All four statements
(E) Question not attempted

32. Consider the following pairs :
1. $f_n(x) = xe^{-nx^2}, x \in \mathbb{R}$: Uniformly convergent on $\mathbb{R}$.
2. $f_n(x) = xe^{-nx^2}, x \in \mathbb{R}$ Uniformly convergent only on compact subset of $\mathbb{R}$.
3. $f_n(x) = xe^{-nx^2}, x \in \mathbb{R}$: Bounded and not uniformly convergent on $\mathbb{R}$.
How many of the above pairs are correctly matched?
(A) Only one pair
(B) Only two pairs
(C) All three pairs
(D) None of the pairs
(E) Question not attempted

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33. Consider the following statements : 1. Let $f_n : [0,1] \to \mathbb{R}$ be a continuous function for each positive integer n. If

$$\lim_{n\to\infty} \int_0^1 f_n(x)^2 dx = 0 \text{ then } \lim_{n\to\infty} f_n(1/2) = 0$$

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2. $\{(x,y) \in \mathbb{R}^2 \mid xy = 1\}$ is connected.
3. $\{(x,y) \in \mathbb{C}^2 \mid xy = 1\}$ is not connected.
4. $\sum_{n=1}^{\infty} x^2 e^{-nx}$ converges uniformly on $(0, \infty)$.
How many of the statements given above are correct?

(A) Only one statement
(B) Only two statements
(C) Only three statements
(D) All four statements
(E) Question not attempted

34. Consider the following statements :
1. Let $\mathbb{N}$ be endowed with the co-finite topology, then $\mathbb{N}$ is not path connected.
2. $GL(n, \mathbb{R})$ has two connected components.
3. $SL(n, \mathbb{R})$ is connected.

How many of the statements given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

35. Consider the following statements :
1. If f is continuous and $-f$ is well defined, then $-f$ is continuous.
2. The union of two locally compact spaces remains locally compact.
3. Total boundness is a topological property.

How many of the statements given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

36. Consider the following Statements :
1. If V_n is open dense subset of $\mathbb{R}, \forall n$, then $\bigcap_{n\geq 1} V_n$ is non empty.
2. For each $n \geq 1, V_n$ is open dense subset of $\mathbb{R}$, then $\bigcap_{n\geq 1} V_n$ is countable. $n \geq 1$
3. For $n \geq 1, V_n$ is open dense subset of $\mathbb{R}$, then $\bigcap_{n\geq 1} V_n$ is uncountable.
4. For $n \geq 1, V_n$ is open dense subset of $\mathbb{R}$ then $\bigcap_{n\geq 1} V_n$ is dense in $\mathbb{R}$.

How many of the statement given above are correct?
(A) Only one statement

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(B) Only two statements
(C) All three statements
(D) All four statements
(E) Question not attempted

37. Given below are two statements :
Statement 1: $f(z) = e^z$ maps vertical line segment $x = a, -\pi < y \leq \pi$ onto circle $|f(z)| = e^a$.
Statement II : $f(z) = e^z$ maps the horizontal line $y = b, -\infty < x < \infty$ onto ray the $\arg(e^z) = b$.
In the light of the above statements, choose the Correct answer from the options given below:
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is correct but Statement II is false
(D) Statement I is incorrect but Statement II is true
(E) Question not attempted

38. Consider the following pairs :
1. $x^3 - 2$: S_3
2. $x^4 - 7$: D_8
3. $x^8 - 24x^6 + 144x^4 - 288x^2 + 144$: Q_8

How many of the above pairs are correctly matched?
(A) Only one pair
(B) Only two pairs
(C) All three pairs
(D) None of the pairs
(E) Question not attempted

39. Given below are two statements :
Statement I : Any order two normal subgroup of a group is a central subgroup.
Statement II : If a p-subgroup of a group is normal then it must be unique.
In the light of the above statements, choose the Correct answer from the options given below :
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is correct but Statement II is false
(D) Statement I is incorrect but Statement II is true
(E) Question not attempted

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40. Given below are two statements :
Statement I : Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(x) \leq f(y)$ for $x \leq y$, then the set where f is not continuous is finite or countably infinite.
Statement II : Let A be a countably infinite subset of $\mathbb{R}$. Then there exists a non-zero continuous function $g : \mathbb{R} \to \mathbb{R}$ such that $\int_a^b g(x)dx = 0$ if $a, b \notin A$.
In the light of the above statements, choose the correct answer from the options given below :
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is correct but Statement II is false
(D) Statement I is incorrect but Statement II is true
(E) Question not attempted

41. Given below are two statements :
Statement I : Let $P_n(x)$ be any polynomial with integer coefficients whose degree is greater than or equal to 1 . Then $\exists$ at least one point $x \in [-2, 2]$ satisfing $|P_n(x)| \geq 2$.
Statement II : Let $P_n(x)$ be any polynomial with real coefficients where degree is greater than or equal to 1 . Then $\exists$ at least one point $x \in [-2, 2]$ satisfing $|P_n(x)| \geq 2$.
In the light of the above statements, choose the correct answer from the options given below :
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is correct but Statement II is false
(D) Statement I is incorreotbut Statement II is true
(E) Question not attempted

42. Consider the following statements :
1. $\mathbb{Z}\left[\frac{1+\sqrt{-3}}{2}\right]$ is a Euclidean domain.
2. $\mathbb{Z}\left[\frac{1+\sqrt{-7}}{2}\right]$ is a Principal Ideal domain.
3. $\mathbb{Z}\left[\frac{1+\sqrt{-23}}{2}\right]$ is a Euctidean domain.

How many of the statement(s) given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None

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(E) Question not attempted

43. Given below are two statements :
Let $\mathbb{Z}[i]$ be the ring of Gaussian integers.
Statement I : $1+10i$ is a unit in $\mathbb{Z}[i]$
Statement II : $3+2i$ is a unit in $\mathbb{Z}[i]$
In the light of the above statements, choose the Correct answer from the options given below :
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is correct but Statement II is false
(D) Statement I is incorreet but Statement II is true
(E) Question not attempted

44. Given below are two statements :

Statement I : Let $f : D(0,2) \to \mathbb{C}$ be analytic function satisfying $f(0) = 1, \left\{ f\left(e^{i\theta}\right) \right| > 2 \forall \theta \in [-\pi, \pi]$. Then $\exists z_0 \in D(0,2) : f(z_0) = 0$.

Statement II : If f is analytic at infinity then derivative of f need not to be analytic at infinity.

In the light of the above statements, choose the Correct answer from the options given below :
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is correct but Statement II is false
(D) Statement I is incorrect but Statement II is true
(E) Question not attempted

45. Given below are two statements :
Statement I : Let f be a non-constant and analytic in an open set containing the closed unit disc. If $|f(z)| = 1$ whenever $|z| = 1$, then for every $w_0 \in \mathbb{C} : |w| < 1, f(z) = w_0$ has a root.
Statement II : Every polynomials of degree n assumes each complex number exactly n times.
In the light of the above statements, choose the Correct answer from the options given below :
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is correct but Statement II is false
(D) Statement I is incorrect but Statement II is true

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(E) Question not attempted

46. Match the number of zeroes for the following functions inside $|z| = 1$.

1. $z^7 - 4z^3 + z + 1$: 3
2. $z^7 + 5z^2 + 12$: 0
3. $e^z - 3z^2$: infinite

How many of the above pairs are correctly matched?

(A) Only one pair
(B) Only two pairs
(C) All three pairs
(D) None of the pairs
(E) Question not attempted

47. Consider the following pairs :

1. $\sum_{n=1}^{\infty} \frac{e^{-n|x|}}{n^3}$: The series converges uniformly on $(-\pi, \pi)$.
2. $\sum_{n=1}^{\infty} \left(\frac{x}{n}\right)^n$: Converges uniformly on $(-\pi, \pi)$.
3. $\sum_{n=1}^{\infty} \frac{1}{[(x+\pi)n]^2}$: Converges uniformly on $(-\pi, \pi)$.

How many of the above pairs are correctly matched?

(A) Only one pair
(B) Only two pairs
(C) All three pairs
(D) None of the pairs
(E) Question not attempted

48. Consider the following pairs :

1. $s_1 = \left\{ \frac{\sin\left(\frac{n\pi}{2017}\right)}{n} : n \in \mathbb{N} \right\}$ s_1 is compact in $\mathbb{R}$.
2. $s_2 = \left\{ \frac{\cos\left(\frac{n\pi}{2017}\right)}{n} : n \in \mathbb{N} \right\}$: s_2 is compact in $\mathbb{R}$.
3. $s_3 = \left\{ \frac{\tan\left(\frac{n\pi}{2017}\right)}{n} : n \in \mathbb{N} \right\}$: s_3 is compact in $\mathbb{R}$.

How many of the above pairs are correctly matched?

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(A) Only one pair
(B) Only two pairs
(C) All three pairs
(D) None of the pairs
(E) Question not attempted

49. Given below are two statements :
Statement I: If a rational function $f(z) = \frac{p(z)}{q(z)}$ has a removable singularity at $z = \infty$ then $\deg p(z) \leq \deg q(z)$.
Statement II : If a rational function $f(z) = \frac{p(z)}{q(z)}$ has a pole (simple pole) at $z = \infty$ then $\deg p(z) > \deg q(z)$.
In the light of the above statements, choose the Correct answer from the options given below:
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is correct but Statement II is false
(D) Statement I is incorrect but Statement II is true
(E) Question not attempted

50. Consider the following pairs :
1. $|z - (1 + 2i)| = \text{Re}(z - 2 + 3i)$: Parabola
2. $|z - (1 + i)| + |z - (-1 + i)| = 2.5$: Ellipse
3. $|z - (1 + i)| = 1$: Circle
How many of the above pairs are correctly matched?
(A) Only one pair
(B) Only two pairs
(C) All three pairs
(D) None of the pairs
(E) Question not attempted

51. Consider the following pairs. Let C be unit circle $\{|z| = 1\}$ with counter clockwise orientation:
1. $\int_C z^m dz = 0$: if $m \neq -1$
2. $\int_C \bar{z}^m dz = 0$: if $m \neq 1$
3. $\int_C z^m |dz| = 0$: if $m \neq 0$

How many of the above pairs are correctly matched?
(A) Only one pair
(B) Only two pairs
(C) All three pairs
(D) None of the pairs
(E) Question not attempted

52. Given below are two statements :

Statement I : Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function satisfying $f(x) = f(x + \sqrt{2}) = f(x + \sqrt{3}) \forall x$. Then f is constant.

Statement II : Let E be a Lebesgue measurable subset of $\mathbb{R}$ with $\mu_L(E) = 1$. Then there exists a Lebesgue mearurable set $A \subset E$ such that $\mu_L(A) = 1/2$.

In the light of the above statements, choose the correct answer from the options given below :
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is correct but Statement II is false
(D) Statement I is incorrect but Statement II is true
(E) Question not attempted

53. Consider the following Statements :
1. Let f be a twice differentiable function on $\mathbb{R}$ such that both f and f'' are strictly positive on $\mathbb{R}$. Then $\lim_{x \to \infty} f(x) = \infty$.
2. Let $f : \mathbb{R}^2 \to \mathbb{R}$ by $f(x, y) = x^{[y]}$, then f is continuous at a point of $\mathbb{R}^2$.
3. There exists a function $f : \mathbb{R} \to \mathbb{R}$ satisfying, $f(-2) = -2, f(2) = 2$ and $|f(x) - f(y)| \leq |x - y|^{3/2} \forall x, y \in \mathbb{R}$.
How many of the statement given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

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54. Given below are two statements :

Statement 1 : Let $f : [-1, 1] \to \mathbb{R}$ be a continuous function. Assume that $f(-1)f(1) < 0$. Then there exists some $\alpha \in (-1, 1)$ such that $f(\alpha) = 0$. (The topology of $\mathbb{R}$ may not be usual topology and the topology of [-1, 1] may not be subspace topology)

Statement II: The pre-image of a connected set by a continuous function is connected.
In the light of the above statements, choose the Correct answer from the options given below :
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is correct but Statement II is false
(D) Statement I is incorrect but Statement II is true
(E) Question not attempted

55. Given below are two statements :

Statement I : $Q^{\mathbb{N}}$ is first countable but not second countable.

Statement II : Let $x_1 = \left\{(x_1, x_2) \in \mathbb{R}^2 \mid 0 < x_1 < \infty, x_2 = \sin \frac{1}{x_1}\right\}$ and $x_2 = \left\{(0, x_2) \in \mathbb{R}^2 \mid -1 \leq x_2 \leq 1\right\}$. Then x_1 and $x_1 \cup x_2$ both are connected, but $x_1 \cup x_2$ is not path-connected.

In the light of the above statements, choose the Correct answer from the options given below :
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is correct But Statement II is false
(D) Statement I is incorrectbut Statement II is true
(E) Question not attempted

56. Given below are two statements:
Statement I: In a discrete probability weighted sum of all the probabilities can be 5.
Statement II: In a discrete probability weighted sum of all the probabilities always 1.
In the light of the above statements, choose the Correct answer from the options given below :
(A) Both statement I and statement II are true.
(B) Both statement I and statement II are false.
(C) Statement I is correct but statement II is false.
(D) Statement I is incorrect but statement II is true.
(E) Question not attempted

57. Given below are two statements:
Statement I: The scatter in a series of values about the average is called central tendency.
Statement II: The scatter in a series of values about the average is called dispersion.
In the light of the above statements, choose the Correct answer from the options given below :
(A) Both statement I and statement II are true.
(B) Both statement I and statement II are false.
(C) Statement I is correct but statement II is false.
(D) Statement I is incorreet but statement II is true.
(E) Question not attempted

58. Consider the following statements:
1. The feasible solution of a LPP belongs to 1st and 2nd quadrants.
2. Only 1st quadrant.
3. Only 1st and 3rd quadrants.
How many of the statements given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

59. Consider the following statements:
1. The maximum ordinate of a normal curve is at $X = \mu$.
2. In a normal curve, $\mu \pm 0.6745\sigma$ covers 50% area.
3. The upper quartiles of a standard normal distribution is 0.6745σ.
How many of the statements given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

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60. Consider the following statements:
1. Poisson distribution is highly skewed with high mean.
2. If X in Poisson, then standard error of X is $\sqrt{\lambda/n}$.
3. In Poisson distribution, parameter $\lambda > 0$ and taking finite value.
How many of the statements given above are correct?
(A) Only one

(B) Only two
(C) All three
(D) None
(E) Question not attempted

61. Considered the following statements:

1. Card $\left(\frac{\mathbb{Z}[i]}{\langle 1+2i\rangle}\right)$ is 2
2. Card $\left(\frac{\mathbb{Z}[i]}{\langle 1+2i\rangle}\right)$ is 5
3. $\frac{\mathrm{Z}[i]}{\langle 1+2i\rangle}$ is a field
4. $\frac{\mathbb{Z}[i]}{\langle 1+2i\rangle}$ has a non-zero proper ideal.

How many of the statements given above are correct?
(A) Only one statement
(B) Only two statements
(C) Only three statements
(D) All four statements
(E) Question not attempted

62. Consider the following pairs:
Let $GL(n, \mathbb{Z}p)$ be the general linear group of $n \times n$ matrices, where $\mathbb{Z}p = \mathbb{F}p$.

1. $o\left(GL\left(2, \mathbb{Z}_2\right)\right)$: 4
2. $o\left(GL\left(2, \mathbb{Z}_2\right)\right)$: 6
3. $o\left(GL\left(3, \mathbb{Z}_2\right)\right)$: 160

How many of the above pairs are correctly matched ?
(A) Only one pair
(B) Only two pairs
(C) All three pairs
(D) None of the pairs
(E) Question not attempted

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63. Consider the following statements:

1. Group of units of the ring $\mathbb{Z}[\sqrt{2}]$ is finite.
2. Group of units of the ring $\mathbb{Z}[\sqrt{2}]$ is infinite.
3. Group of units of the ring $\mathbb{Z}[\sqrt{-2}]$ is finite.
4. Group of units of the fing $\mathbb{Z}[\sqrt{-10}]$ has finite order.

How many of the statements given above are correct?

(A) Only one statement
(B) Only two statements
(C) Only three statements
(D) All four statements
(E) Question not attempted

64. Given below are two statements :
Statement I : If G_1 and G_2 are two groups such that $G_1 \subseteq G_2$, then G_1 is a subgroup of G_2.
Statement II: If G_1 and G_2 are two subgroups of a group G, then $G_1 \cup G_2$ is a subgroup of G_1.
In the light of the above statements, choose the Correct answer from the options given below
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is correct but Statement II is false
(D) Statement I is incorrect but Statement II is true
(E) Question not attempted

65. Given below are two statements :

Let P be a prime number and $f(x) \in \mathbb{Z}[x]$ with $\deg(f(x)) \geq 1$. Let $\bar{f}(x) \in \mathbb{F}_P[x]$ then :

Statement I : If $\bar{f}(x)$ is irreducible over $\mathbb{F}_p$ then $f(x)$ is irreducible over Q.

Statement II : If $\bar{f}(x)$ is reducible over $\mathbb{F}_p$ then $f(x)$ is reducible over Q.

In the light of the above statements, choose the Correct answer from the options given below :
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is correct but Statement II is false
(D) Statement I is incorrect but Statement II is true
(E) Question not attempted

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66. Consider the following Statements :
1. $x^4 + 3x^4 + 6x + 3$ is irreducible over Q.
2. $10x^5 + 11x^3 + 121x + 11$ is reducible over Q.
3. $x^3 + 3x^2 + 2x + 1$ is irreducible over Q.
4. $x^3 + 4x^2 + x + 1$ is irredacible over Q.

How many of the statements given above are correct?
(A) Only one statement
(B) Only two statements

(C) Only three statements
(D) All four statements
(E) Question not attempted

67. Give below are two statements: Let $I = \int_a^b x^2 dx$; a and b are given.
Statement I: The value of I obtained by using trapezoidal rule is always greater than or equal to the exact value of definite integral.
Statement II: The value of 1 obtained by using Simpson's rule is always equal to exact value of the definite integral.
In the light of the above statements, choose the correct answer from the options given below :
(A) Both statement I and statement II are true.
(B) Both statement I and statement II are false.
(C) Statement I is correct but statement II is false.
(D) Statement I is incorrect but statement II is true.
(E) Question not attempted

68. Consider the following pairs: Match the order
1. Bisection method: 1
2. Secant method: Golden Ratio
3. Newton Raphson method: 2
How many of the above pairs are correctly matched?
(A) Only one pair
(B) Only two pairs
(C) All three pairs
(D) None of the pairs
(E) Question not attempted

69. Given below are two statements:
Statement I: Two steps of Euler's method with step size 0.25 give same accuracy to one step of modified Euler's method with step size 0.5 .
Statement II: Two steps of Euler's method with step size 0.1 give more accurate solution than one step of modified Euler's method with step size 0.2 .
In the light of the above statements, choose the Correct answer from the options given below :
(A) Both statement I and statement II are true.
(B) Both statement I and statement II are false.
(C) Statement I is correct But statement II is false.
(D) Statement I is incorrect but statement II is true.
(E) Question not attempted

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70. Given below are two statements :
Statement I: If $o(G) = 175$, then there exists a subgroup H of G such that $o(H) = 35$.
Statement II: The number of non-isomorphic abelian groups of order 24 is equal to 4 . In the light of the above statements, choose the Correct answer from the options given below :
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is correct but Statement II is false
(D) Statement I is incorrect but Statement II is true
(E) Question not attempted

71. Consider the following statements: Let G be a p-group.
1. G can not be simple.
2. G may be simple.
3. If $o(G) = p^n$ then G is not simple for any $n \in \mathbb{N} - \{1\}$.
How many of the statements given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

72. Consider the following pairs, Let G be a group and p be a prime number
1. $o(G) = p^2$: Abelian group
2. $o(G) = p$: Cyclic group
3. $o(G) = p^3$: May be abelian
How many of the above pairs are correctly matched?
(A) Only one pair
(B) Only two pairs
(C) All three pairs
(D) None of the pairs
(E) Question not attempted

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73. Given below are two statements:

Statement 1: The pth root of a given number N is root of equation $f(x) : x^p - N = 0$ then iteration formula by Newton Raphson method is: $x_{n+1} = \dfrac{(p-1)x_n^p + N}{p{x_n}^{p-1}}$

Statement II: The iteration formula for $f(x) : x^p - N = 0$ by Secant method is: $x_{n+1} = x_n - \dfrac{x_n^p - N}{x_n^p - x_{n-1}^p}(x_n - x_{n-1})$.

In the light of the above statements, choose the Correct answer from the options given below :
(A) Both statement I and statement II are true.
(B) Both statement I and statement II are false.
(C) Statement I is correct but statement II is false.
(D) Statement I is incorrect but statement II is true.
(E) Question not attempted

74. Consider the following pairs: Match the order of fitting polynomial
1. Trapezoidal rule : 1
2. Simpson's $\frac{1}{3}$ rule : 2
3. Simpson's $\frac{3}{8}$ rule : **3**

How many of the above pairs are correctly matched?
(A) Only one pair
(B) Only two pairs
(C) All three pairs
(D) None of the pairs
(E) Question not attempted

75. Consider the following statements :
Let R be the ring of all continuous functions $f : \mathbb{R} \to \mathbb{R}$ and $B = \{f \in R \backslash f(0) = \text{even integer }\}$. Then
1. B is a subring of R.
2. B is an ideal of R.
3. B is an integral domain.

How many of the statements given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

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76. Given below are two statements :
Statement I : Let G be an abelian group with $o(G) = 122$, then there are 62 elements of order k, if $1 \leq k < 122$.
Statement II: If $o(G) = 2809$, then G is abelian.
In the light of the above statements, choose the Correct answer from the options given below :
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false

(C) Statement I is correct but Statement II is false
(D) Statement I is incorrect but Statement II is true
(E) Question not attempted

77. Given below are two statements :
Let G be a group and H be a subgroup of G. Then
Statement I : G/H is a group of cosets of H, where $G/H = \{aH \mid aHbH = abH\}$ and H is any subgroup of G.
Statement II: G/H is a group for some subgroup H.
In the light of the above statements, choose the Correct answer from the options given below :
(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is correct but Statement II is false
(D) Statement I is incorrect but Statement II is true
(E) Question not attempted

78. Consider the following statements:
1. A continuous random variable assume any possible value between two points.
2. A discrete random variable assume only finite values.
3. A discrete sample space has only finite points.
4. Any random value can be assumed by any random variable.
How many of the statements given above are correct?
(A) Only one statement
(B) Only two statements
(C) Only three statements
(D) All four statements
(E) Question not attempted

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79. Consider the following statements and the options given below: Let x_i represent equally spaced points with $x_i - x_{i-1} = n$. Then difference formula for approximation of $f'(x_i)$ is.

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1. $\dfrac{f(x_{i+1}) - f(x_i)}{n}$
2. $\dfrac{f(x_i) - f(x_{i-1})}{2n}$
3. $\dfrac{f(x_{i+1}) - f(x_{i-1})}{2n}$
4. $\dfrac{f(x_i) - f(x_{i-1})}{n}$

How many of the options given above are correct?
(A) Only one
(B) Only two
(C) Only three
(D) All four
(E) Question not attempted

80. Give below are two statements: Let f be differentiable on $[-1,1]$. Let $f(-1) = Ef(1), f(0) = -1$.
Statement I: $\exists c \in (-1,1)$ such that $f(c) = 0$.
Statement II: $\exists c \in (-1,1)$ such that $f'(c) = 0$.
In the light of the above statements, choose the Correct answer from the options given below :
(A) Both statement I and statement II are true.
(B) Both statement I and statement II are false.
(C) Statement I is correct but statement II is false.
(D) Statement I is incorrect but statement II is true.
(E) Question not attempted

81. For $\lambda, \mu \in \mathbf{R}$, consider the following integral equation

$$f(x) = e^x + \lambda \int_0^1 e^{i\mu(x-t)} f(t)dt, x \in \mathbb{R}$$

Consider the following statements.
Statement I: There is a solution to the above integral equation for any choice of λ, $\mu \in \mathbf{R}$.
Statement II: The solution to the above integral equation does not exist if $\lambda = 1$ and $\mu = 3$.
In the light of the above statements, choose the correct answer from the options given below:
(A) Statement I is correct but statement II is incorrect.
(B) Statement I is incorrect but statement II is correct.
(C) Both Statement I and statement II are correct.
(D) Neither Statement I nor statement II are correct.
(E) Question not attempted

82. For $\lambda \in \mathbb{R}$ and $K : [0,1] \times [0,1] \to \mathbb{R}$ be a continuously differentiable function, consider the following integral equation

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$$f(x) = \sin\left(x^2\right) + \lambda \int_0^x K(x,y) f(y) dy, x \in [0,1]$$

Now consider the following statements.
Statement I : The above integral equation has a unique solution if and only if $|\lambda| < 1$.
Statement II: The above integral equation has a unique solution for any choice of $\lambda \in \mathbb{R}$ and a continuously differentiable function $K : [0,1] \times [0,1] \to \mathbb{R}$.
In the light of the above statements, choose the Correct answer from the options given below
(A) Statement I is correct but statement II is incorrect.
(B) Statement I is incorrec but statement II is correct.
(C) Both Statement I and statement II are correct.
(D) Neither Statement I nor statement II are correct.
(E) Question not attempted

83. Assume that $p, q : |a,b| \to \mathbb{R}$ be two continuous functions. Now for the following ODE

$$y''(x) + p(x)y'(x) + q(x)y(x) = 0, x \in (x,b)$$

Consider the following statements:
Statement-I : Any solution to the above ODE whose graph is tangent to x-axis at some point $c \in (a,b)$ is identically zero.
Statement-II : Any solution to the above ODE whose graph has tangent not equal to the x -axis but parallel to x -axis at some point $d \in (a,b)$ is identically zero.
In the light of the above statements, choose the Correct answer from the options given below:
(A) Statement-I is correct but statement-II is incorrect.
(B) Statement-I is incorrect but statement-II is correct.
(C) Both statement-I and statement-II are correct.
(D) Neither statement-I nor statement-II is correct.
(E) Question not attempted

84. Consider the following statements:
Statement-I : The initial value problem

$$\frac{dy}{dx} = y^{4/3}, y(0) = 1$$

has a unique solution which exists for each $x \geq 0$.
Statement-II : The maximum interval of existence for solution to the initial value problem

$$\frac{dy}{dx} = y^{4/3} y(0) = 1$$

is $\mathbb{R}$.
In the light of the above statements, choose the Correct answer from the options given below:
(A) Statement-I is correct but statement-II is incorrect.
(B) Statement-I is incorrect but statement-II is correct.
(C) Both statement-I and statement-II are correct.
(D) Neither statement-I nor statement-II is correct.
(E) Question not attempted

85. Consider the following functional

$$J\left(y_1(x), y_2(x)\right) = \int_0^1 \left[y_1^2(x) - 2y_1y_2(x) + \left(\frac{dy_1}{dx}\right)^2 + \left(\frac{dy_2}{dx}\right)^2\right] dt$$

Statement I : The Euler-Lagrange equation corresponding to the functional $J\left(y_1(x), y_2(x)\right)$ is a system of linear differential equations.
Statement II: If $y_1(0) = y_2(0) = 0$, the there exists a unique solution to the EulerLagrange equation corresponding to the functional $J\left(y_1(x), y_2(x)\right)$.
Statement III : If $y_1(0) = y_2(1) = 0$, the there exists a unique solution to the EulerLagrange equation corresponding to the functional $J\left(y_1(x), y_2(x)\right)$.
Statement IV : There is no solution to the Euler-Lagrange equation corresponding to the functional $J\left(y_1(x), y_2(x)\right)$ satisfying $y_1(0) = y_2(0) = 0$ and $y_1'(0) = y_2'(0) = 1$.
How many of the statements given above are correct?
(A) Only one statement
(B) Only two statements
(C) Only three statements
(D) All four statements
(E) Question not attempted

86. For $a, b \in \mathbb{R}$ with $a < b$ and $\lambda \in \mathbb{R}$, consider the following integral equation

$$f(x) = e^x + \lambda \int_a^b \sin(2y) f(y) dy, x \in [a, b]$$

Now consider the following statements.
Statement I : There exist a unique pair $a, b \in \mathbb{R}$ with $a < b$ such that the above integral equation has a unique solution for any choice of λ.
Statement II : Given $a, \lambda \in \mathbb{R}$, the above integral equation can not have a unique solution for any choices of $a, b \in \mathbb{R}$ with $a < b$.
Statement III : Given $a, b \in \mathbb{R}$ with $a < b$, there exists a unique $\lambda \in \mathbb{R}$ such that the above integral equation has a unique solution.

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Statement IV : There alwaysexists a choice of $a < b$ and $\lambda \in \mathbb{R}\backslash\{0\}$ for which the above integral equation has a unique solution.

How many of the statements given above are correct?

(A) Only one statement
(B) Only two statements
(C) Only three statements
(D) All four statements
(E) Question not attempted

87. Consider the following statements

Statement-I : The initial value problem

$$y' = 1 + y^{2/3}, y(0) = 0$$

has a unique solution in some open interval around 0 .

Statement-II : There exists a solution of $3|y'| + 2|y| + 1 = 0$.

Statement-III: There exists a solution to the initial value problem

$$y' = -y^2, y(1) = 1$$

whose interval of existence contains some open interval around 0 .

Statement-IV : There exists is a nonzero solution to the initial value problem

$$y' = -y^2, y(0) = 0$$

whose interval of existence is $\mathbb{R}$.

How many of the statements given above are correct?

(A) Only one statement
(B) Only two statements
(C) Only three statements
(D) All four statements
(E) Question not attempted

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88. For the following partial differential equation (PDE)

$$\frac{\partial^2 u}{dx^2} - 4\frac{\partial^2 u}{\partial x \partial y} + 4\frac{\partial^2 u}{\partial y^2} = 0$$

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Consider the following statements.

Statement I: The above PDE is elliptic.

Statement II: For any two fimes differentiable functions $f, g : \mathbb{R} \to \mathbb{R}$ the function

$$v(x, y) := f(y + 2x) + xg(x + 2y), (x, y) \in \mathbb{R}^2$$

is a solution to the above PDE.

In the light of the above statements, choose the correct answer from the options given below:

(A) Statement I is correct but statement II is incorrect.
(B) Statement I is incorrect, but statement II is correct.
(C) Both Statement I and statement II are correct.
(D) Neither Statement I nor statement II are correct.
(E) Question not attempted

89. Consider the following statements.
Statement I: There exists a unique solution to the Euler-Lagrange equation corresponding to the functional

$$J(y) := \int_0^{100} \left(y^2 + \frac{dy}{dx}\right) dx, y(0) = 0, y(100) = 1$$

Statement II: There exists a unique solution to the Euler-Lagrange equation corresponding to the functional

$$J(y) := \int_0^{100} y^2(x)dx, y(0) = 0, y(100) = 1$$

Statement III : The solution y to the Euler-Lagrange equation corresponding to the functional

$$J(y) := \int_0^{100} \left(y^2 + \frac{dy}{dx}\right) dx, y(0) = 0, y(100) = 1$$

is a polynomial of degree 2.
How many of the statements given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

90. Consider the following statements.
Statement I : There exists a nonzero characteristic $\lambda \in \mathbb{R}$, for the following integral equation

$$f(x) = \tan x + \lambda \int_0^x \cos^2(y) f(y) dy, x \in \left[0, \frac{\pi}{4}\right]$$

Statement II : For any $\lambda \neq 0$, there exists a nonzero solution to the following integral equation

$$f(x) = \lambda \int_0^x \cos^2(y) f(y) dy, x \in [0, \pi/4]$$

such that $f(0) = 0$.

Statement III : All continuous solutions to the above integral equations are Lipschitz continuous on $\left[0, \frac{\pi}{4}\right]$.

How many of the statements given above are correct?

(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

91. Assume that p, q are continuous functions on $[a, b]$ and y_1 and y_2 be two solutions of $\frac{d^2y}{dx^2} + p(x)\frac{dy}{dx} + q(x)y = 0$, on (a, b) are such that their Wronskian and the solution y_1 at some $x_0 \in (a, b)$ are zero. Now consider the following statements:

Statement-I : $y_1(x) = 0$, for all $x \in (a, b)$.

Statement-II: $y_2(x) = \left(\frac{y_2'(x_0)}{y_1'(x_0)}\right) y_1(x)$, for all $x \in (a, b)$.

In the light of the above statements, choose the options given below which is always correct.

(A) Statement-I is correct but statement-II is incorrect
(B) Statement-I is incorrect but statement-II is correct
(C) Either statement-I is correct or statement-II is correct
(D) Neither statement-I nor statement-II is correct
(E) Question not attempted

92. Consider the following second order linear ordinary differential equation:

$$\begin{cases} x^2y''(x) + y(x) = 0, 1 \leq x \leq a, \\ y(1) = y(a) = 0. \end{cases}$$

Suppose
$A := \{a < 1 : (1) \text{ has a nonzero solution }\}$.

Statement-I : $A = \phi$.

Statement-II : A is countably infinite set.

In the light of the above statements, choose the Correct answer from the options given below :

(A) Statement-I is correct but statement-II is incorrect.
(B) Statement-I is incorrect but statement-II is correct.
(C) Both statement-I and statement-II are correct.
(D) Neither statement-I nor statement-II is correct.

(E) Question not attempted

93. For the following partial differential equation (PDE)

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \text{ in } \mathbb{R}^2$$

Consider the following statements.

Statement I: There exists a non zero solution u to the above PDE such that

$$\lim_{\left(\sqrt{x^2+y^2}\right)\to\infty} u(x,y) \text{ exists finitely.}$$

Statement II: If u is a solution to the above PDE then
$v(x,y) := u(y,-3x); (x,y) \in \mathbb{R}^2$ is also a solution to the above PDE.

Statement III: There are infinitely many nonconstant smooth functions which solve the above PDE.

How many of the statements given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

94. Consider the following Cauchy problem for first order partial differential equation (PDE)

$$y\frac{\partial u}{\partial x} + x\frac{\partial u}{\partial y} = 0, (x,y) \in \mathbb{R} \times (0,\infty), u(x,0) = 0, x \in \mathbb{R}$$

Now consider the following statements.
Statement I: There exists a unique solution to the above problem which exists on $\mathbb{R} \times [0,\infty)$.
Statement II: There exists a nonzero solution u to the above problem on $\mathbb{R} \times [0,\infty)$ such that the set $A := \{(x,y) \in \mathbb{R} \times [0,\infty) : u(x,y) = 0\}$ is uncountable.
Statement III: The solution to the above problem does not exist on $\mathbb{R} \times [0,\infty)$.
How many of the statements given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

95. Consider the following statements:

1. Angle between the incident and scattered direction is called solid angle.
2. Halley's comet has a value of eccentricity 0.096 .
3. The velocity of satellite is $\sqrt{2gR^2\left(\frac{1}{r}-\frac{1}{2a}\right)}$.

How many of the statements given above are correct?

(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

96. Consider the following pairs:

1. $\frac{d}{dt}\left(\frac{\partial \vec{r_i}}{\partial q_j}\right) : -\frac{\partial}{\partial q_j}\left(\frac{d\vec{r_i}}{dt}\right)$
2. $\frac{d}{dt}\left(\frac{\partial \vec{r_i}}{\partial q_j}\right) : \frac{\partial}{\partial q_j}\left(\frac{d\vec{r_i}}{dt}\right)$
3. $\frac{d}{dt}\left(\frac{\partial \vec{r_i}}{\partial q_j}\right) : \frac{d}{dq_j}\left(\frac{d\vec{r_i}}{dt}\right)$

How many of the above pairs are correctly matched?

(A) Only one pair
(D) Only two pairs
(C) All three pairs
(D) None of the pairs
(E) Question not attempted

97. Consider the following statements.

Statement I: There exists a non zero continuous function $f : [a, b] \to \mathbb{R}$ such that $\int_a^b f(x)g(x)dx = 0$, for any choice of continuous function $g : (a, b) \to \mathrm{R}$ such that $g(a) = g(b) = 0$.

Statement II: There exists atleast one non zero continuous functions f_1 or $f_2 : [a, b] \to \mathbb{R}$ such that $\int_a^b [f_1(x)g_1(x) + f_2(x)g_2(x)]\,dx = 0$, for any choice of continuous functions $g_1, g_2 : (a, b) \to \mathbb{R}$ such that $g_1(a) = g_1(b) = g_2(a) = g_2(b) = 0$.

In the light of the above statements, choose the correct answer from the options given below:

(A) Statement I is correct but statement II is incorrect.
(B) Statement I is incorrect but statement II is correct.

(C) Both Statement I and statement II are correct.
(D) Neither Statement I nor statement II are correct.
(E) Question not attempted

98. Consider the following statements.

Statement I: The Euler-Lagrange equation corresponding to the functional $J(u) = \int_\Omega \left(1 + |\nabla u(x)|^2\right)^{1/2} dx$ is a nonlinear partial differential equation where $\Omega \subset \mathbb{R}^n$ is an open set and $u : \mathbb{R}^n \to \mathbb{R}$ is a two times continuously differentiable function.

Statement II: There exists a solution to the Euler-Lagrange equation corresponding to the functional $J(u) := \int_\Omega \left(1 + |\nabla u(x)|^2\right)^{1/2} dx$ where $\Omega \subset \mathbb{R}^n$ is an open set and $u : \mathbb{R}^n \to \mathbb{R}$ is a two times continuously differentiable function.

In the light of the above statements, choose the correct answer from the options given below:
(A) Statement I is correct but statement II is incorrect.
(B) Statement I is incorrectobut statement II is correct.
(C) Both Statement I and statement II are correct.
(D) Neither Statement I nor statement II are correct.
(E) Question not attempted

99. Suppose $g : R \to R$ is a continuously differentiable function such that $g'(t) \geq 0$ for all $t \in \mathbb{R}$. Now if $u, v : [a, b] \to \mathbb{R}$ are continuous functions such that

$$u(t) < 10 + \int_a^t g(u(s))ds \text{ and}$$

$$v(t) > 10 + \int_a^t g(v(s))ds$$

hold for any $t \in [a, b]$. Now consider the following statements.
Statement-I : $u(t) < v(t)$ for all $t \in [a, b]$.
Statement-II : $u(t) > v(t)$ for all $t \in [a, b]$.
Statement-III : $u(t) = v(t)$ for all $t \in [a, b]$.
How many of the statements given above are correct?
(A) Only one
(B) Only two
(C) All three
(D) None
(E) Question not attempted

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100. Consider the following statements
Statement-I : $\lambda = 0$, is an eigenvalue of

$$y'' + \lambda y = 0, y(0) = y(\pi) + ky'(\pi) = 0$$

if and only if $k = -\pi$.
Statement-II : There are finitely many solution to

$$y' = 5y, y(0) = 0$$

but infinitely many solution to

$$y' = y^{1/4}, y(0) = 0$$

Statement-III : There are exactly two solutions to

$$y'' + y = 0, y(0) = 0, y(\pi) = 1$$

How many of the statements given above are correct?
(A) Only one
(D) Only two
(C) All three
(D) None
(E) Question not attempted

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[ANSWER KEY]

Q. No.	Ans	Q. No.	Ans	Q. No.	Ans	Q. No.	Ans	Q. No.	Ans
1	C	2	A	3	B	4	A	5	C
6	B	7	D	8	A	9	B	10	C
11	B	12	B	13	B	14	C	15	A
16	B	17	B	18	D	19	B	20	C
21	A	22	B	23	A	24	D	25	B
26	A	27	D	28	B	29	C	30	C
31	D	32	A	33	A	34	C	35	D
36	C	37	A	38	C	39	C	40	C
41	C	42	B	43	A	44	C	45	A
46	B	47	B	48	B	49	A	50	C
51	C	52	A	53	D	54	B	55	D
56	C	57	D	58	A	59	C	60	B
61	B	62	A	63	C	64	B	65	B
66	C	67	A	68	C	69	C	70	C
71	B	72	C	73	A	74	C	75	A
76	A	77	D	78	A	79	C	80	A
81	B	82	B	83	A	84	B	85	A
86	A	87	A	88	D	89	D	90	A
91	C	92	B	93	B	94	B	95	D
96	A	97	D	98	C	99	A	100	A

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